Accuracy Limits of Causal Trees for Individualized Treatment Effects

CART-type recursive partitioning, small cells, and pointwise reliability

Matias D. Cattaneo
Princeton University

Jason M. Klusowski
Princeton University

Ruiqi (Rae) Yu
Princeton University

June 2026

Introduction and Overview

Main Question

When can adaptive causal trees be trusted pointwise?

Causal trees are attractive because they search for interpretable subgroups and estimate treatment effects within terminal nodes.

The core difficulty: greedy CART-type splitting can create terminal nodes with very few observations with nonvanishing probability.

The statistical target is local:

$$\tau(\mathbf{x}) = \mathbb{E}[y_i(1) - y_i(0) \mid \mathbf{x}_i = \mathbf{x}].$$

Individualized and subgroup decisions need accuracy at specific covariate values, not just good averages.

This paper: even in a randomized experiment with constant CATE, standard causal trees can converge arbitrarily slowly in uniform norm.

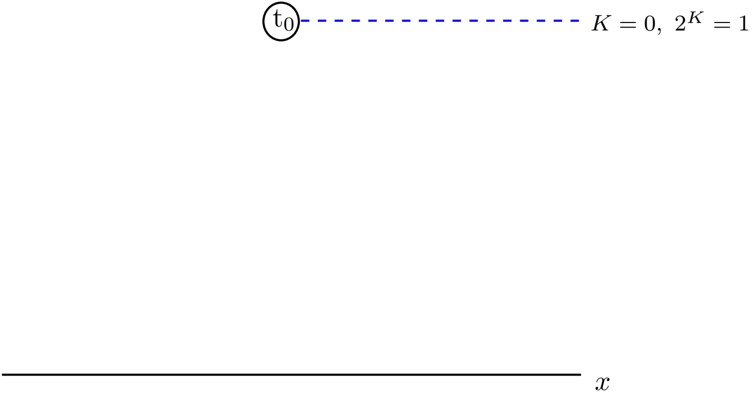
Small cells create variance, not smoothing bias.

Three Takeaways

1. **Uniform and pointwise reliability can fail**: CART-type causal trees may converge more slowly than any polynomial rate in n.
2. **Honesty is not enough**: sample splitting removes a slowly varying factor, but it does not remove the small-cell mechanism.
3. **Average accuracy can still look good**: integrated mean squared error can be near-parametric in the constant-effect benchmark.

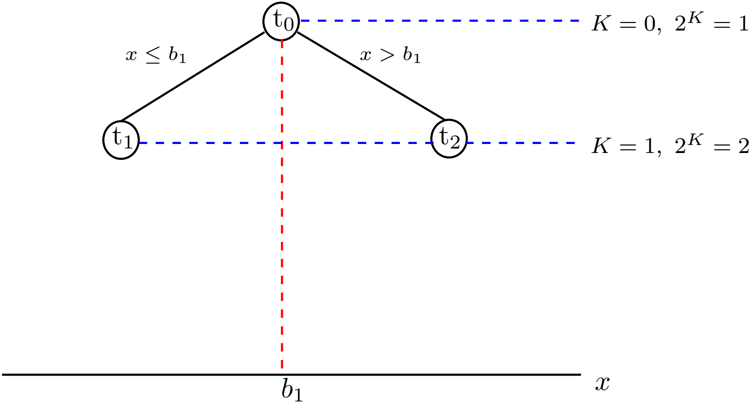
Implication: regularization, balance restrictions, and inference procedures should be analyzed as part of the estimator, not as implementation details.

Adaptive Axis-Aligned Decision Tree (CART)



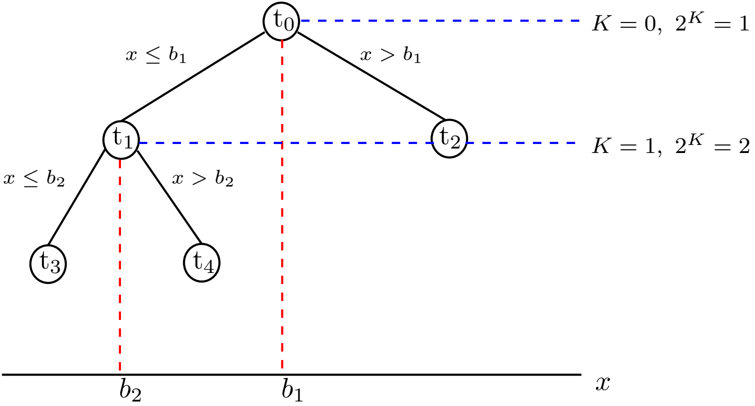
for each K : $$\min_{j=1,2,\cdots,p} \min_{\beta_1,\beta_2,\tau} \sum_{\mathbf{x}_i \in \mathrm{t}} \Big(y_i - \beta_1 \mathbb{1}(x_{ij} \le \tau) - \beta_2 \mathbb{1}(x_{ij} > \tau) \Big)^2$$

Adaptive Axis-Aligned Decision Tree (CART)



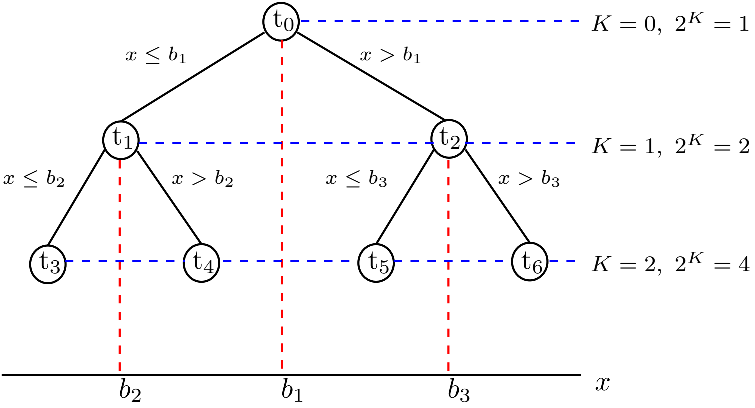
for each K : $$\min_{j=1,2,\cdots,p} \min_{\beta_1,\beta_2,\tau} \sum_{\mathbf{x}_i \in \mathrm{t}} \Big(y_i - \beta_1 \mathbb{1}(x_{ij} \leq \tau) - \beta_2 \mathbb{1}(x_{ij} > \tau) \Big)^2$$

Adaptive Axis-Aligned Decision Tree (CART)



for each K : $$\min_{j=1,2,\cdots,p} \min_{\beta_1,\beta_2,\tau} \sum_{\mathbf{x}_i \in \mathrm{t}} \Big(y_i - \beta_1 \mathbb{1}(x_{ij} \leq \tau) - \beta_2 \mathbb{1}(x_{ij} > \tau) \Big)^2$$

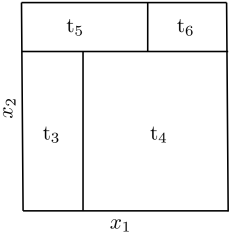
Adaptive Axis-Aligned Decision Tree (CART)

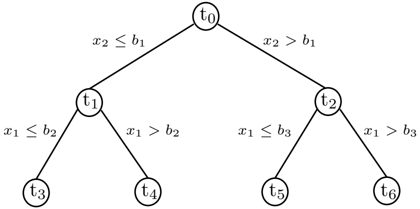


for each K :

$$\min_{j=1,2,\cdots,p} \min_{\beta_1,\beta_2,\tau} \sum_{\mathbf{x}_i \in \mathrm{t}} \Big(y_i - \beta_1 \mathbb{1}(x_{ij} \le \tau) - \beta_2 \mathbb{1}(x_{ij} > \tau) \Big)^2$$

Adaptive Axis-Aligned Decision Tree (CART vs. "Honesty")



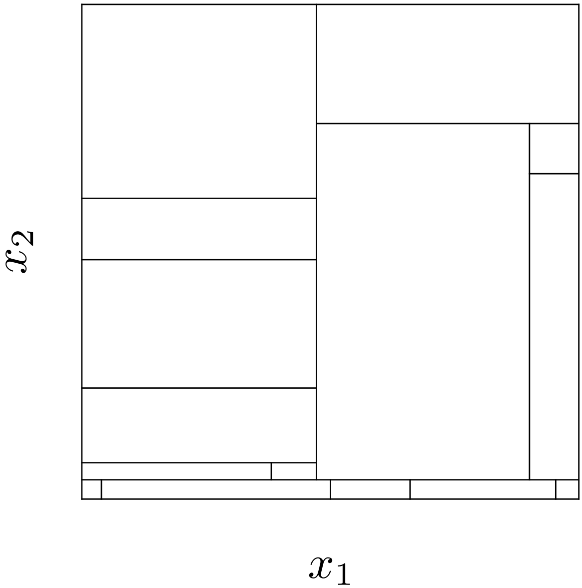


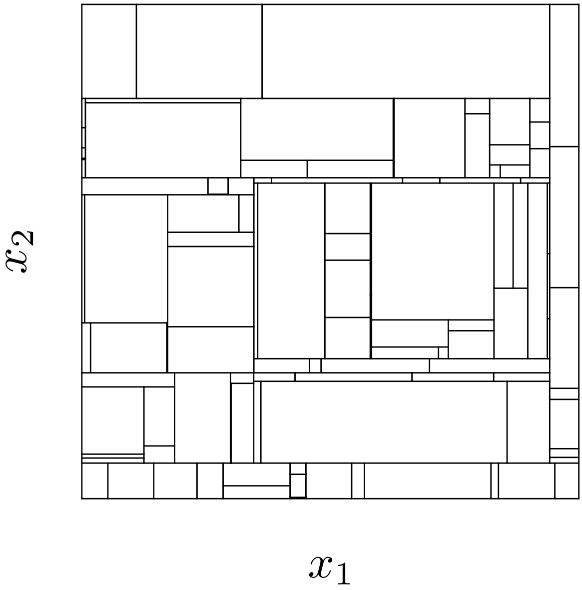
- Full sample (or CART):

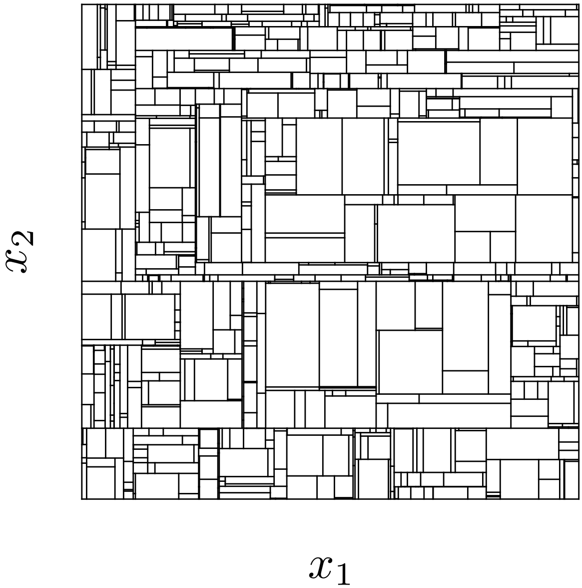
$$\widehat{\mu}(\mathbf{x}; \mathsf{T}_K) = \frac{1}{n(\mathsf{t})} \sum_{\mathbf{x}_i \in \mathsf{t}} y_i, \qquad n(\mathsf{t}) = \sum_{\mathbf{x}_i \in \mathsf{t}} \mathbb{1}(\mathbf{x}_i \in \mathsf{t})$$

- Sample splitting (or "honesty"): $\mathsf{T}_K \perp\!\!\!\perp (\widetilde{y}_i, \widetilde{\mathbf{x}}_i : i = 1, \ldots, n_\mu)$, and

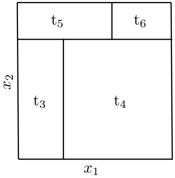
$$\check{\mu}(\mathbf{x}; \mathsf{T}_K) = \frac{1}{\widetilde{n}(\mathsf{t})} \sum_{\widetilde{\mathbf{x}}_i \in \mathsf{t}} \widetilde{y}_i, \qquad \widetilde{n}(\mathsf{t}) = \sum_{\widetilde{\mathbf{x}}_i \in \mathsf{t}} \mathbb{1}(\widetilde{\mathbf{x}}_i \in \mathsf{t}).$$

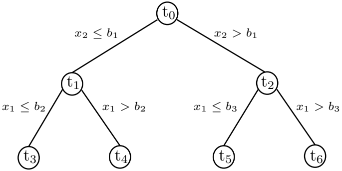

x_2
x_1


x_2
x_1


x_2
x_1

Adaptive Axis-Aligned Decision Tree (CART)





Full sample: $$\widehat{\mu}(\mathbf{x};\mathsf{T}_K) = \frac{1}{n(\mathsf{t})}\sum_{\mathbf{x}_i\in\mathsf{t}} y_i, \qquad n(\mathsf{t}) = \sum_{\mathbf{x}_i\in\mathsf{t}} \mathbb{1}(\mathbf{x}_i \in \mathsf{t})$$

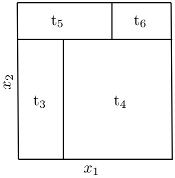
"Honesty": $$\check{\mu}(\mathbf{x};\mathsf{T}_K) = \frac{1}{\widetilde{n}(\mathsf{t})}\sum_{\widetilde{\mathbf{x}}_i\in\mathsf{t}} \widetilde{y}_i, \qquad \widetilde{n}(\mathsf{t}) = \sum_{\widetilde{\mathbf{x}}_i\in\mathsf{t}} \mathbb{1}(\widetilde{\mathbf{x}}_i \in \mathsf{t}).$$

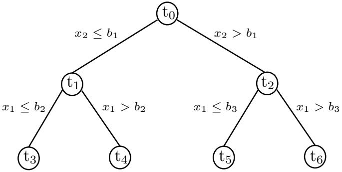
Uniform Result: for $\mu(\mathbf{x}) = \mu$, and for all $K \geq 1$ and $b \in (0,1)$,

$$\liminf_{n\to\infty} \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathcal{X}} \big|\widehat{\mu}(\mathbf{x};\mathsf{T}_K) - \mu(\mathbf{x})\big|^2 \geq C_1 \frac{\log\log(n)}{n^b}\Big) \geq C_2 b,$$

$$\liminf_{n\to\infty} \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathcal{X}} \big|\check{\mu}(\mathbf{x};\mathsf{T}_K) - \mu(\mathbf{x})\big|^2 \geq C_3 \frac{1}{n^b}\Big) \geq C_4 b,$$

Adaptive Axis-Aligned Decision Tree (CART)





Full sample: $$\widehat{\mu}(\mathbf{x};\mathsf{T}_K) = \frac{1}{n(\mathsf{t})}\sum_{\mathbf{x}_i\in\mathsf{t}} y_i, \qquad n(\mathsf{t}) = \sum_{\mathbf{x}_i\in\mathsf{t}} \mathbb{1}(\mathbf{x}_i\in\mathsf{t})$$

"Honesty": $$\check{\mu}(\mathbf{x};\mathsf{T}_K) = \frac{1}{\widetilde{n}(\mathsf{t})}\sum_{\widetilde{\mathbf{x}}_i\in\mathsf{t}} \widetilde{y}_i, \qquad \widetilde{n}(\mathsf{t}) = \sum_{\widetilde{\mathbf{x}}_i\in\mathsf{t}} \mathbb{1}(\widetilde{\mathbf{x}}_i\in\mathsf{t}).$$

Mean Square Result: for $\mu(\mathbf{x}) = \mu$, and for all $K \geq 1$ and $b \in (0,1)$,

$$\mathbb{E}\Big[\int_{\mathcal{X}} \big|\widehat{\mu}(\mathbf{x};\mathsf{T}_K) - \mu(\mathbf{x})\big|^2 dF_{\mathbf{X}}(\mathbf{x})\Big] \leq C_1 \frac{2^K \log^5(n)}{n} + \frac{2^K \log^4(n)\log(p)}{n},$$

$$\mathbb{E}\Big[\int_{\mathcal{X}} \big|\check{\mu}(\mathbf{x};\mathsf{T}_K) - \mu(\mathbf{x})\big|^2 dF_{\mathbf{X}}(\mathbf{x})\Big] \leq C_2 \frac{2^K \log^5(n)}{n},$$

Causal Decision Trees

Causal Trees in the Literature

The revised paper studies a broad CART-type mechanism

PNAS

Recursive partitioning for heterogeneous causal effects

Susan Athey[a,1] and Guido Imbens[a]

[a]Stanford Graduate School of Business, Stanford University, Stanford, CA 94305

Edited by Richard M. Shiffrin, Indiana University, Bloomington, IN, and approved May 20, 2016 (received for review June 25, 2015)

In this paper we propose methods for estimating heterogeneity in causal effects in experimental and observational studies and for conducting hypothesis tests about the magnitude of differences in treatment effects across subsets of the population. We provide a data-driven approach to partition the data into subpopulations

Within the prediction-based machine learning literature, regression trees differ from most other methods in that they produce a partition of the population according to covariates, whereby all units in a partition receive the same prediction. In this paper, we focus on the analogous goal of deriving a partition

Common motivation: let the data discover treatment-effect subgroups while keeping the resulting leaf estimates interpretable.

Common design choices: transformed outcomes, treated-control contrasts, local squared-error splitting, and sample splitting.

Our focus: the statistical consequences of canonical greedy recursive partitioning, not a single article or software implementation.

Causal Decision Trees

PNAS

Recursive partitioning for heterogeneous causal effects

Susan Athey[a,1] and Guido Imbens[a]

[a]Stanford Graduate School of Business, Stanford University, Stanford, CA 94305

Edited by Richard M. Shiffrin, Indiana University, Bloomington, IN, and approved May 20, 2016 (received for review June 25, 2015)

In this paper we propose methods for estimating heterogeneity in causal effects in experimental and observational studies and for conducting hypothesis tests about the magnitude of differences in treatment effects across subsets of the population. We provide a data-driven approach to partition the data into subpopulations

Within the prediction-based machine learning literature, regression trees differ from most other methods in that they produce a partition of the population according to covariates, whereby all units in a partition receive the same prediction. In this paper, we focus on the analogous goal of deriving a partition

"It enables researchers to let the data discover relevant subgroups while preserving the validity of confidence intervals constructed on treatment effects within subgroups."

"Honesty has the implication that the asymptotic properties of treatment effect estimates within the partitions are the same as if the partition had been exogenously given. Although there is a loss of precision due to sample splitting (which reduces sample size in each step of estimation), there is a benefit in terms of eliminating bias that offsets at least part of the cost."

Setup

- DGP: $\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \ldots, n\}$ i.i.d., and $y_i = y_i(1) \cdot d_i + y_i(0) \cdot (1 - d_i)$.
- RCT: $(y_i(0), y_i(1), \mathbf{x}_i^\top) \perp\!\!\!\perp d_i$ and $\xi = \mathbb{P}(d_i = 1) \in (0, 1)$.
- CATE: $\tau_{\texttt{CATE}}(\mathbf{x}_i) = \mathbb{E}[y_i(1) - y_i(0) \mid \mathbf{x}_i = \mathbf{x}]$.
- Identification:

$$\tau_{\texttt{CATE}}(\mathbf{x}_i) = \mathbb{E}\Big[y_i \frac{d_i - \xi}{\xi(1 - \xi)} \mid \mathbf{x}_i\Big] = \mathbb{E}[y_i \mid \mathbf{x}_i,\ d_i = 1] - \mathbb{E}[y_i \mid \mathbf{x}_i,\ d_i = 0].$$

Causal Trees Methodology:

1. CATE estimator: IPW vs. DIM at terminal nodes (final partition).
2. Tree construction: (adaptive) recursive partitioning with causal-inference-tailored splitting criteria.
3. Data usage: full-sample vs. "honesty".

CATE Estimator

Suppose T is the tree used, and $\mathcal{D}_\tau = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \ldots, n_\tau\}$, with $n_\tau \leq n$, is the dataset used. Let t be the unique terminal node in T containing $\mathbf{x} \in \mathcal{X}$.

- The *Inverse Probability Weighting* (IPW) estimator is

$$\widehat{\tau}_{\mathtt{IPW}}(\mathbf{x}; \mathsf{T}, \mathcal{D}_\tau) = \frac{1}{n(\mathsf{t})} \sum_{i:\mathbf{x}_i \in \mathsf{t}} \frac{d_i - \xi}{\xi(1-\xi)} y_i,$$

where $n(\mathsf{t}) = \sum_{i=1}^{n_\tau} \mathbb{1}(\mathbf{x}_i \in \mathsf{t})$ is the "local" sample size.

- The *Difference-in-Means* (DIM) estimator is

$$\widehat{\tau}_{\mathtt{DIM}}(\mathbf{x}; \mathsf{T}, \mathcal{D}_\tau) = \frac{1}{n_1(\mathsf{t})} \sum_{i:\mathbf{x}_i \in \mathsf{t}} d_i y_i - \frac{1}{n_0(\mathsf{t})} \sum_{i:\mathbf{x}_i \in \mathsf{t}} (1 - d_i) y_i,$$

where $n_d(\mathsf{t}) = \sum_{i=1}^{n_\tau} \mathbb{1}(\mathbf{x}_i \in \mathsf{t},\ d_i = d)$, for $d = 0, 1$, are the "local" sample sizes.

Tree Construction

Suppose $\mathcal{D}_\mathsf{T} = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \ldots, n_\mathsf{T}\}$, with $n_\mathsf{T} \leq n$, is the dataset used to construct the tree T. There is a unique node $\mathrm{t}_0 = \mathcal{X}$ at initialization, and child nodes are generated by iterative axis-aligned splitting of the parent node based on either of the following two rules.

- *Variance Maximization*: A parent node t (i.e., a terminal node partitioning $\mathcal{X}$) in a previous tree T' is divided into two child nodes, t_L and t_R, forming the new tree T, by maximizing
$$\frac{n(\mathrm{t}_\mathrm{L})n(\mathrm{t}_\mathrm{R})}{n(\mathrm{t})}\Big(\widehat{\tau}_l(\mathrm{t}_\mathrm{L}; \mathsf{T}, \mathcal{D}_\mathsf{T}) - \widehat{\tau}_l(\mathrm{t}_\mathrm{R}; \mathsf{T}, \mathcal{D}_\mathsf{T})\Big)^2, \qquad l \in \{\mathtt{DIM}, \mathtt{IPW}\}.$$
The two final causal trees are denoted by $\mathsf{T}_{\mathtt{DIM}}(\mathcal{D}_\mathsf{T})$ and $\mathsf{T}_{\mathtt{IPW}}(\mathcal{D}_\mathsf{T})$, respectively.

- *SSE Minimization*: A parent node t (i.e., a terminal node partitioning $\mathcal{X}$) in the previous tree T' is divided into two child nodes, t_L and t_R, forming the next tree T, by solving
$$\min_{a_\mathrm{L}, b_\mathrm{L}, a_\mathrm{R}, b_\mathrm{R} \in \mathbb{R}} \sum_{\mathbf{x}_i \in \mathrm{t}_\mathrm{L}} (y_i - a_\mathrm{L} - b_\mathrm{L} d_i)^2 + \sum_{\mathbf{x}_i \in \mathrm{t}_\mathrm{R}} (y_i - a_\mathrm{R} - b_\mathrm{R} d_i)^2,$$
where only the data $\mathcal{D}_\mathsf{T}$ is used.
The final causal tree is denoted by $\mathsf{T}_{\mathtt{SSE}}(\mathcal{D}_\mathsf{T})$.

Data Usage

Recall that $\mathcal{D} = \{(y_i, d_i, \mathbf{x}_i^\top) : i = 1, 2, \ldots, n\}$ is the available random sample.

- *Full sample*: The dataset $\mathcal{D}$ is used for both tree construction and treatment effect estimation. The causal tree estimators are

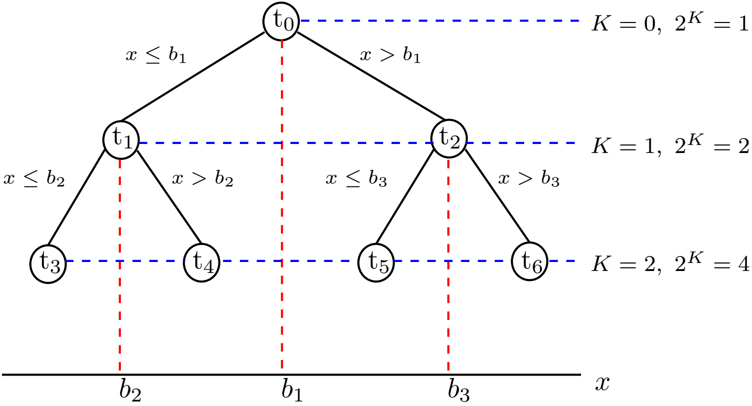
$$\widehat{\tau}_{\mathtt{IPW}}(\mathbf{x}) = \widehat{\tau}_{\mathtt{IPW}}(\mathbf{x}; \mathsf{T}_{\mathtt{IPW}}(\mathcal{D}), \mathcal{D}),$$
$$\widehat{\tau}_{\mathtt{DIM}}(\mathbf{x}) = \widehat{\tau}_{\mathtt{DIM}}(\mathbf{x}; \mathsf{T}_{\mathtt{DIM}}(\mathcal{D}), \mathcal{D}),$$
$$\widehat{\tau}_{\mathtt{SSE}}(\mathbf{x}) = \widehat{\tau}_{\mathtt{DIM}}(\mathbf{x}; \mathsf{T}_{\mathtt{SSE}}(\mathcal{D}), \mathcal{D}).$$

- *Honesty*: The dataset $\mathcal{D}$ is divided into independent datasets $\mathcal{D}_{\mathsf{T}}$ and $\mathcal{D}_{\tau}$ for tree construction and treatment effect estimation. The causal tree estimators are

$$\check{\tau}_{\mathtt{IPW}}(\mathbf{x}) = \widehat{\tau}_{\mathtt{IPW}}(\mathbf{x}; \mathsf{T}_{\mathtt{IPW}}(\mathcal{D}_{\mathsf{T}}), \mathcal{D}_{\tau}),$$
$$\check{\tau}_{\mathtt{DIM}}(\mathbf{x}) = \widehat{\tau}_{\mathtt{DIM}}(\mathbf{x}; \mathsf{T}_{\mathtt{DIM}}(\mathcal{D}_{\mathsf{T}}), \mathcal{D}_{\tau}),$$
$$\check{\tau}_{\mathtt{SSE}}(\mathbf{x}) = \widehat{\tau}_{\mathtt{DIM}}(\mathbf{x}; \mathsf{T}_{\mathtt{SSE}}(\mathcal{D}_{\mathsf{T}}), \mathcal{D}_{\tau}).$$

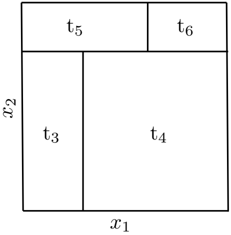
Main Results

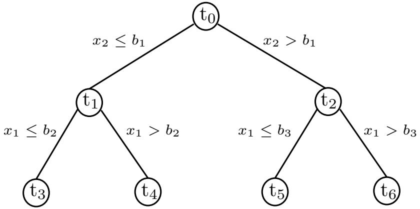
Adaptive Axis-Aligned Decision Tree (CART)



for each K : $$\min_{j=1,2,\cdots,p} \min_{\beta_1,\beta_2,\tau} \sum_{\mathbf{x}_i \in \mathrm{t}} \Big(y_i - \beta_1 \mathbb{1}(x_{ij} \le \tau) - \beta_2 \mathbb{1}(x_{ij} > \tau) \Big)^2$$

Adaptive Axis-Aligned Decision Tree (CART vs. "Honesty")





- Full sample (or CART):

$$\widehat{\mu}(\mathbf{x}; \mathsf{T}_K) = \frac{1}{n(\mathsf{t})} \sum_{\mathbf{x}_i \in \mathsf{t}} y_i, \qquad n(\mathsf{t}) = \sum_{\mathbf{x}_i \in \mathsf{t}} \mathbb{1}(\mathbf{x}_i \in \mathsf{t})$$

- Sample splitting (or "honesty"): $\mathsf{T}_K \perp\!\!\!\perp (\widetilde{y}_i, \widetilde{\mathbf{x}}_i : i = 1, \ldots, n_\mu)$, and

$$\check{\mu}(\mathbf{x}; \mathsf{T}_K) = \frac{1}{\widetilde{n}(\mathsf{t})} \sum_{\widetilde{\mathbf{x}}_i \in \mathsf{t}} \widetilde{y}_i, \qquad \widetilde{n}(\mathsf{t}) = \sum_{\widetilde{\mathbf{x}}_i \in \mathsf{t}} \mathbb{1}(\widetilde{\mathbf{x}}_i \in \mathsf{t}).$$

Benchmark for the Lower Bounds

$$y_i(d) = c_d + \varepsilon_i(d), \qquad d \in \{0, 1\}, \qquad \tau(\mathbf{x}) = c_1 - c_0 \equiv \tau.$$

Assumption (Favorable Causal DGP)

1. $(y_i(0), y_i(1), d_i, \mathbf{x}_i^\top)$, $i = 1, \ldots, n$, *is a random sample.*
2. *Treatment is randomized:* $(y_i(0), y_i(1), \mathbf{x}_i^\top) \perp\!\!\!\perp d_i$ *and* $\mathbb{P}(d_i = 1) = \xi \in (0, 1)$.
3. *The CATE is constant over* $\mathcal{X} \subseteq \mathbb{R}^p$.
4. *Covariates are independent and continuously distributed; by monotone transformation, take* $\mathbf{x}_i \sim \mathsf{Unif}([0, 1]^p)$.
5. *Potential-outcome errors are light-tailed with positive variance.*

The benchmark removes confounding, treatment-effect complexity, and smoothing bias. Any failure comes from the tree construction and within-leaf averaging.

Decision Stumps

For each level K, adaptive (CART) decision trees solve

$$\min_{j=1,2,\cdots,p} \min_{\beta_1,\beta_2,\tau} \sum_{\mathbf{x}_i \in \mathsf{t}} \Big(y_i - \beta_1 \mathbb{1}(x_{ij} \le \tau) - \beta_2 \mathbb{1}(x_{ij} > \tau) \Big)^2,$$

which is equivalent to maximizing the so-called *impurity gain*

$$\sum_{\mathbf{x}_l \in \mathsf{t}} (y_l - \mu)^2 - \sum_{\mathbf{x}_l \in \mathsf{t}} \big(y_l - \overline{y}_{\mathsf{t}_L} \mathbb{1}(x_{lj} \le \tau) - \overline{y}_{\mathsf{t}_R} \mathbb{1}(x_{lj} > \tau) \big)^2$$

$$= \frac{1}{i(n(\mathsf{t}) - i)} \Big(\frac{1}{\sqrt{n(\mathsf{t})}} \sum_{l=1}^{i} (y_l - \mu) - \frac{i}{n(\mathsf{t})} \frac{1}{\sqrt{n(\mathsf{t})}} \sum_{l=1}^{n(\mathsf{t})} (y_l - \mu) \Big)^2$$

with respect to index i and variable j, after reordering the data $\implies (\hat{\imath}, \hat{\jmath})$.

- Darling-Erdös (1956) limit law (Berkes & Weber, 2006): for any non-decreasing function $1 \le h(m) \le m$ for which $\lim_{m \to \infty} h(m) = \infty$ and any $w \in \mathbb{R}$,

$$\mathbb{P}\left(\max_{m/h(m) \le i \le m} \left| \frac{1}{\sqrt{i}} \sum_{l=1}^{i} (y_l - \mu) \right| < \lambda(h(m), w) \right) \to e^{-e^{-w}},$$

as $m \to \infty$, where $\lambda(\cdot, \cdot)$ is known.

Decision Stumps: Split Location and Covariate Selected

Careful study of maximum over different ranges of the split index gives:

Theorem

For each $a, b \in (0,1)$ *with* $a < b$ *and* $j \in \{1, 2, \ldots, p\}$,

$$\liminf_{n\to\infty} \mathbb{P}\big(n^a \le \hat{\imath} \le n^b,\ \ \hat{\jmath} = j\big) = \liminf_{n\to\infty} \mathbb{P}\big(n - n^b \le \hat{\imath} \le n - n^a,\ \ \hat{\jmath} = j\big) \ge \frac{b-a}{2pe}.$$

- With positive probability, split index $\hat{\imath}$ concentrates near extremes.
- Too few observations will be available on one of the cells after the first split for CART to deliver a polynomial-in-n consistent estimator of μ.
- Decision stumps exhibit slower than any polynomial-in-n convergence rate.
- Intuitively, iterating the procedure down the tree T_K can only make things worse.

Main Theorems: Accuracy Contrast

Six CART-type causal tree estimators, $l \in \{\texttt{IPW}, \texttt{DIM}, \texttt{SSE}\}$

Let Assumption DGP hold and let the tree depth be at most K.

Uniform accuracy can be arbitrarily slow

For every $b \in (0,1)$, there are positive constants independent of n and K such that

$$\liminf_{n\to\infty} \mathbb{P}\left(\sup_{\mathbf{x}\in\mathcal{X}} |\widehat{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 \geq C_1 \frac{\log\log(n)}{n^b}\right) \geq C_2 b,$$
$$\liminf_{n\to\infty} \mathbb{P}\left(\sup_{\mathbf{x}\in\mathcal{X}} |\check{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 \geq C_3 \frac{1}{n^b}\right) \geq C_4 b.$$

Average accuracy can still be small

$$\mathbb{E}\left[\int_{\mathcal{X}} |\widehat{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 \, dF_{\mathbf{X}}(\mathbf{x})\right] \leq C_1 \frac{2^K \log^4(n)\log(np)}{n},$$
$$\mathbb{E}\left[\int_{\mathcal{X}} |\check{\tau}_l(\mathbf{x}) - \tau(\mathbf{x})|^2 \, dF_{\mathbf{X}}(\mathbf{x})\right] \leq C_2 \frac{2^K \log^5(n)}{n}.$$

Discussion and Simulations

Interpreting the Result

What the lower bound is, and is not, saying

Not a bias story. The constant-CATE benchmark removes smoothing and misspecification bias.

Not fixed by honesty. Independent estimation data does not prevent the selected partition from containing small cells.

Not just one split. Deeper trees inherit and refine the small-cell regions created upstream.

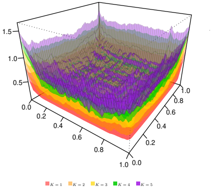
Average-vs-local tension. Integrated MSE can be small while the estimator remains unreliable at some covariate values.

Regularization matters. Minimum node sizes and balance restrictions can limit small cells, but change the estimator and tuning tradeoffs.

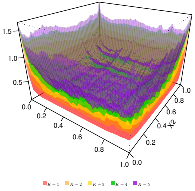
Forest theory connection. Positive guarantees often impose balance restrictions absent from canonical CART-type splitting.

Monte Carlo Evidence ($p = 2$)

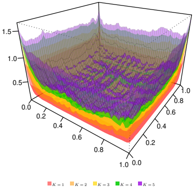
Pointwise RMSE for full-sample estimators $\widehat{\tau}_l$ and honest estimators $\check{\tau}_l$



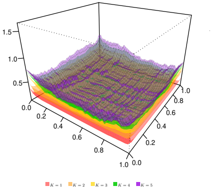
(a) $\widehat{\tau}_{\mathtt{IPW}}$



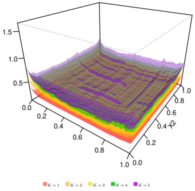
(b) $\widehat{\tau}_{\mathtt{DIM}}$



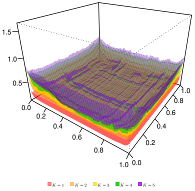
(c) $\widehat{\tau}_{\mathtt{SSE}}$



(d) $\check{\tau}_{\mathtt{IPW}}$



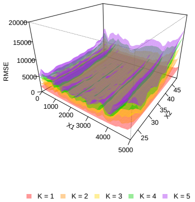
(e) $\check{\tau}_{\mathtt{DIM}}$



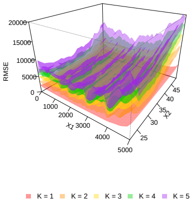
(f) $\check{\tau}_{\mathtt{SSE}}$

Empirical Resampling Evidence: JTPA ($p = 2$)

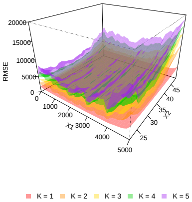
The same small-cell pattern appears on empirical covariate support



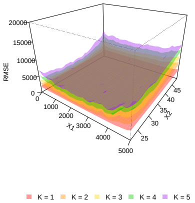
(a) $\widehat{\tau}_{\mathtt{IPW}}$



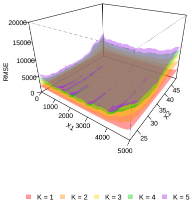
(b) $\widehat{\tau}_{\mathtt{DIM}}$



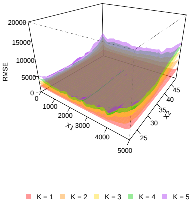
(c) $\widehat{\tau}_{\mathtt{SSE}}$



(d) $\check{\tau}_{\mathtt{IPW}}$



(e) $\check{\tau}_{\mathtt{DIM}}$



(f) $\check{\tau}_{\mathtt{SSE}}$

Conclusion

Conclusion

CART-type causal trees are useful, but their local accuracy has intrinsic limits.

1. **Worst-case pointwise error can decay more slowly than any polynomial rate**, even with randomized treatment and constant CATE.
2. **Honesty does not eliminate the problem**: the partition can still contain very small terminal cells.
3. **Average performance can be reassuring but incomplete**: integrated MSE can be small while individualized or subgroup estimates remain unreliable.

Regularization, balance restrictions, pruning, and inference must be analyzed as part of the procedure. They are not harmless implementation details.